

Centripetal Force

Equipment:

INCLUDED:

Qty	Components	Model #	Purpose
1	Centripetal Force Apparatus	ME-8088	
1	Force Sensor	PS-2104	Measure Centripetal Force
1	Photogate Head	ME-9498A	Measure Speed
1	Large Rod Base	ME-8735	Level the Base
1	90 cm Steel Rod	ME-8738	Fix Height of Force Sensor
1	Multi-Clamp	SE-9507	
1	45 cm Steel Rod	ME-8736	Fix Horizontal Position of Force Sensor
1	Banana Plug Cord-Red (5 pack)	SE-9750	
1	550 Universal Interface	UI-5001	Necessary Hardware
1	PASCO Capstone	UI-5400	Software

Introduction:

In this activity, you will use a Force Sensor and Photogate to discover the relationship of centripetal force, mass, (tangential) speed and radius for an object in uniform circular motion. You will determine what happens to centripetal force as the result of changes in mass, speed, and radius.

Theory:

According to Newton's First Law, an object in motion tends to stay in motion in a straight line at a constant speed if there is no external net force applied to the object. An object undergoing uniform circular motion (motion in a circle at constant speed) must be acted on by a non-zero net force. That net force is called the centripetal force. It must point toward the center of the circle and have a constant magnitude given by:

$$F_c = mv^2/r \quad \text{Eq. (1)}$$

Where m is the mass of the object moving in a circle, r is the radius of the circle, and v is the (tangential) speed of the object. In *procedure* r of this experiment, you will change the radius and the relation between r and F should be described by:

$$F_c = m\omega^2 r \quad \text{Eq. (2)}$$

where ω is angular velocity.

It is important to note that the centripetal force is not an additional force on the system, but rather the vector sum of the forces acting on the object.

$$\mathbf{F}_c = \Sigma \mathbf{F} \quad \text{Eq. (3)}$$

Examples of centripetal force include the tension in a string attached to a can twirled in a circular path, the friction between the road and the tires of a car on an unbanked curve, or the force of gravity pulling a satellite toward the center of Earth as the satellite moves in a circular orbit.

Setup:

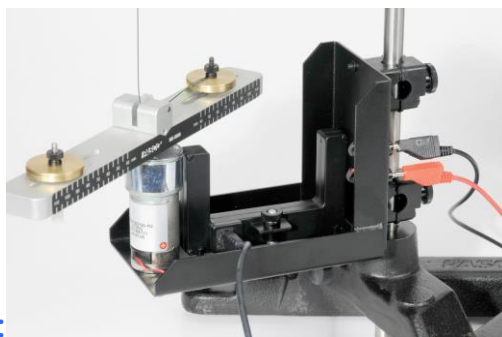


Figure 2a: Complete Setup

Figure 1: Photogate Attachment

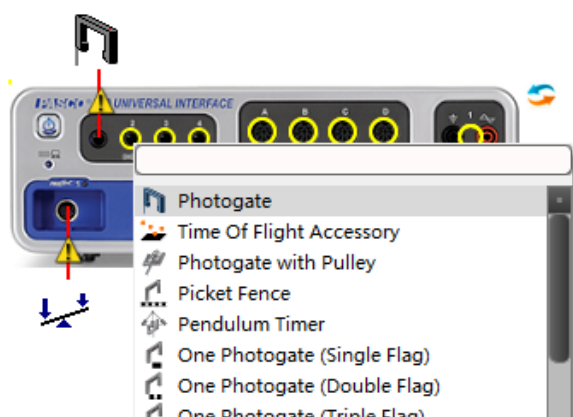


Figure 2b: Photogate Setup

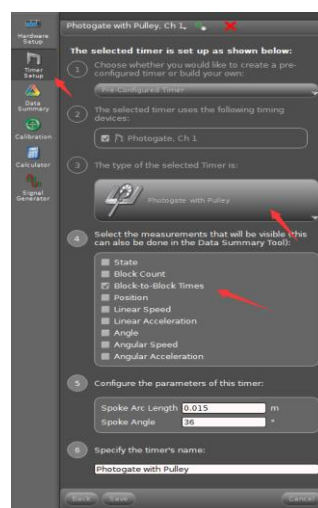


Figure 2c: Timer Setup

1. Attach the Photogate to the frame of the Centripetal Force Apparatus as shown in Figure1.

2. Attach the entire Centripetal Force Apparatus as low as possible to the 90 cm rod and base.
3. Attach the 45 cm rod horizontally to the 90 cm rod with the multi-clamp.
4. Hang the Force Sensor from the horizontal rod.
5. Screw the Ball Bearing Swivel to the Force Sensor.
6. Use a metal clip to attach the low-stretch cable to the Swivel. Thread the other end of the cable through the plastic pulley and attach it to the sliding post by putting the loop over the post and attaching the mass above it. Note: between uses, it is important to store the cable in a manner that does not put kinks in it.
7. Plug the Photogate into Digital Channel 1 on the 550 Universal Interface. Plug the Force Sensor into any *Pasport* input. Here you need setup the Photogate in software. Right-click and then choose *Photogate* like the Figure 2b. Setup step by step of *Timer Setup* as Figure 2c.
8. Click open the Signal Generator at the left of the screen. On 550 Output 1, the Waveform should be DC with a DC Voltage of 4.0 V. Click the Off button to be sure the Signal Generator is off and connect the Centripetal Force Apparatus to OUTPUT 1 on the 550 Universal Interface with banana plugs. It is not important which plug on the Centripetal Force Apparatus attaches to the red output although it will affect the rotation direction. **Careful: if the Signal Generator is not off, the arm will begin to rotate!** Click the Signal Generator again to close it.
9. (This step can be skipped) Level the base. Remove the counterweight mass and put a level on the rotating arm as shown in Figure 3. Start with the arm parallel to the two leveling screws in the base as shown in Figure 3. There is some play in the rotating arm. Tip it up or down a bit by pushing on the end of the rotating arm away from the level. Adjust the leveling screws until the bubble moves about the same amount on either side of center when you rock the arm up and down as in Figures 4a & 4b. Now rotate the arm 90° and repeat changing both screws by the same amount. Then rotate back to the original position and re-level if necessary.
10. Attach a 5 g counterweight mass.
11. Remove the assembly that holds the moving mass (black plastic nut and bolt, silver nut and two plastic washers) and determine its mass. Click open the Calculator at the left of the screen and replace the 0.0038 value in line 7 your measured mass (in kg) for "m hold".
12. Re-attach the assembly to the rotating arm with one plastic washer below the arm and one above. Then the silver nut, tightened down enough to prevent tipping, but the assembly must slide freely. Then the cable loop (see Figure 5). Then use the black nut to attach 5 g of mass above the loop. The cable loop should be **below** the 5 g mass. **It is shown incorrectly in Figures 1&2.**



Figure 3: Level Placement



Figure 4a & 4b: Rocking the Bubble

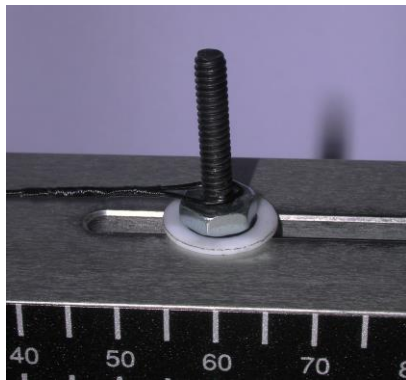


Figure 5: Slide Assembly

Setup C:

1. Now, adjust the height of the force sensor so that the 5g mass is about 10.0 cm from the center when there is just enough tension in the cable to straighten the cable. It is important that the force sensor be exactly above the center of the apparatus. To check this, pull on the mass to put tension in the cable. Then observe the cable from the front of the system to verify it is parallel to the 90 cm rod and measure the distance from the rod to the top of the string and to just above the pulley. Adjust the 45 cm rod holding the force sensor until the cable is parallel to the rod. Note that any time you change the position of the Force Sensor, you will have to repeat this step.
2. Pull the mass to tighten the cable to determine the actual radius to the nearest 0.1 cm. Although you can do this using the scale attached to the side of the apparatus, it is more precise to measure from the center of the mass to where the vertical cable is with a small ruler. Click open the Calculator at the left of the screen. In line 1, replace the 0.1 value (10 cm) with the value that you actually measured. Click the Calculator again to close it.

Procedure m – Force vs Mass (Radius and Velocity held Constant)

1. Press the “ZERO” button on the Force Sensor. This should set the Force Sensor to zero since there should be no force on it at this point.
2. Click RECORD. Collect data for about 5 s and then click STOP to stop data collection. Mean speed and Mean Force should both be zero. If the Mean Force is not zero, record the value. It will help during the data analysis.
3. To avoid wobbling, tighten an identical mass (5 g here) to the opposite side of the rotating platform as a counterbalance so that it is as far from the center of the rotating platform as the distance you measured in step 2. This measurement need not be precise and it is sufficient to use the scale on the side of the apparatus.
4. Click on the Signal Generator at the left of the screen. Under the Output 1, set waveform to DC and the DC Voltage to 4.5 V. **Caution! The next step will cause the apparatus to begin rotation. Be sure it can do so without hitting anything or anybody.** Click the On button. After about 10 s, reduce the voltage to 4.0 V. Starting with a higher speed and then reducing it minimizes the frictional effects by positioning the mass at about the correct position before reducing the speed. Let it run for 20 to 60 s to reach constant speed. (The speed may decrease continually, run until the speed decrease slowly.)
5. Press the RECORD button. Allow data collection to occur for approximately 10 seconds until the mean Values for Speed and Force are almost constant. Press the STOP button.
6. Click the Signal Generator Output 1 to Off. The rotation should stop.
7. In row 1 (5g row) of the Variable Mass table, enter the value of the Mean Speed into column 2 and record the Mean Force (ignore the minus sign) in column 3.
8. Increase the mass by 5.0 g (0.005 kg).
9. Repeat steps 3-8 until you reach 30.0g of mass. Note that the mean speed should be about the same for all the runs.

Analysis m: Force vs Mass

1. Observe the Force vs Mass Graph. The “Av Force” is the measured average force from the table on the previous page.
2. Click open the Calculator at the left of the screen and examine lines 1-7 to verify that “theory force” is calculated using Equation 1 from Theory. Note that the actual measured speed, “Av speed” from the Variable Mass table on the previous page is used to calculate

“theory force”. Since the speed is not quite constant, the “theory force” points are not quite linear. Click the Calculator to close it.

3. The mass sometimes hangs up a little causing an obviously bad “Av Force” point. You should repeat the run for any bad point and see if it improves. Enter the new run in the row below the 0.030 kg row of the Variable Mass table on the previous page.
4. Select the “Av Force” data by clicking in the Legend box or on a data point. Then click the black triangle by the Curve Fit tool in the graph toolbox and select Linear.

Procedure v: Force vs Speed (Radius and Mass held Constant)

1. Keep the **30 g** mass attached to the cable and the rotating platform as it was in Procedure m.
2. Click on the Signal Generator at the left of the screen. Under the Output 1, set the DC Voltage to 5.5 V. **Caution! The next step will cause the apparatus to begin rotation. Be sure it can do so without hitting anything or anybody.** Click the On button. After about 10 s, reduce the voltage to 5.0 V. Let it run for about 20 s to reach constant speed.
3. Press the RECORD button. Allow data collection to occur for approximately 10 seconds until the mean Values for Speed and Force are almost constant. Press the STOP button.
4. In row 1 (5.0 V row) of the Variable Mass table, enter the value of the Mean Speed into column 2 and record the Mean Force (ignore the minus sign) in column 3.
5. Decrease the Signal Generator voltage by 0.5 V. Let it run for about 20 s to reach constant speed.
6. Repeat steps 3-5 until you reach 3.5 V.

Analysis v: Force vs Speed

1. Observe the Force vs Speed² Graph. The “Ave F” is the measured average force from the table on the previous page.
2. Click open the Calculator at the left of the screen and examine line 8 to verify that “theory f” is calculated using Equation 1 from Theory. Click the Calculator to close it.
3. The mass sometimes hangs up a little causing an obviously bad “Ave F” point. You should repeat the run for any bad point and see if it improves. Enter the new run in the row below the 3.5 V row of the Variable Speed table on the previous page.
4. Select the “Ave F” data by clicking in the Legend box or on a data point. Then click the black triangle by the Curve Fit tool in the graph toolbox and select Linear.

Procedure r: Force vs Radius (Mass and Time for one rotation held Constant)

1. Keep the **30 g** mass attached to the cable and the rotating platform as it was in Procedure v.
2. Enter the radius of the circle in column 1 of the Variable Radius table. Note that the initial radius is the same as before and is recorded in line 1 of the Calculator. After you enter it, the value should also show in the Radius box in the upper left.
3. Note: Re-enter the radius every time once you change the radius
4. Click on the Signal Generator at the left of the screen. Under the Output 1, set the DC Voltage to 5.0 V. **Caution! The next step will cause the apparatus to begin rotation. Be sure it can do so without hitting anything or anybody.** Click the On button. After about 10 s, reduce the voltage to 4.5 V. Let it run for about 20 s to reach constant speed.
5. Press the RECORD button. Allow data collection to occur for approximately 10 seconds until the mean Values for Speed and Force are almost constant. Press the STOP button.
6. In row 1 of the Variable Radius table, enter the value of the Mean Speed into column 2 and record the Mean Force (ignore the minus sign) in column 3.
7. Turn the Signal Generator Off. The rotation should stop.
8. Following the steps in Setup C, decrease the radius to about 8.5 cm. Measure the value of the radius to within 0.1 cm. Move the counterweight mass to about the same distance. Repeat steps 2-6.
9. Continue as above for radii of about 7 cm and 5 cm.

Analysis r: Force vs Circle Radius

1. Observe the Force vs Radius, r , Graph. The “Ave Force” is the measured average force from the table on the previous page.
2. Click open the Calculator at the left of the screen and examine lines 9 & 10 to verify that “theo f ” is calculated using Equation 1 from Theory. Note that the actual measured speed, “Ave speed” from the Variable Mass table on the previous page is used to calculate “theo f ”. Since the speed is not quite constant, the “theo f ” points are not quite linear. Click the Calculator to close it.

3. The mass sometimes hangs up a little causing an obviously bad “Ave Force” point. You should repeat the run for any bad point and see if it improves. Enter the new run in the Variable Mass table on the previous page.
4. Select the “Ave Force” data by clicking in the Legend box or on a data point. Then click the black triangle by the Curve Fit tool in the graph toolbox and select Linear.

Conclusions:

1. Looking at the graph on the “Analysis m” tab, what can you conclude about the relationship between the centripetal force and mass? Explain how you know.
2. Looking at the graph on the “Analysis v” tab, what can you conclude about the relationship between the centripetal force and tangential speed? Explain how you know.
3. Looking at the graph on the “Analysis r” tab, what can you conclude about the relationship between the centripetal force and radius of the circle? Explain how you know.
4. Considering the answers to the first three questions, is Equation 1 from Theory valid. Careful...how can the relationship you saw in Question 3 agree with Equation 1?
5. How would friction between the mass and the rotating arm affect your values for the measured force?